

Tianxin Xie

✉ [Email](#) | ☎ [+14372698167](#) | 🐙 [GitHub](#) | 🌐 [Website](#) |  [LinkedIn](#)

SKILLS

AI & ML: PyTorch, LangChain, TensorFlow.js, OpenCV, Sentence-Transformers, FAISS, RAG pipelines

Languages: Python, Kotlin, C++, JavaScript, TypeScript, Java, Verilog, HTML, CSS, Assembly

Tools & Platforms: Git, AWS, KLayout, Virtuoso, Claude Code, Android Studio, VS Code, JetBrains

EXPERIENCE

Software Engineer – Integrated Circuits (IC) Imaging

May. 2026 - Aug. 2026

LOWP Technology Inc.

Markham, Ontario

- Cross-validated CNN- and OpenCV-generated masks' conflicting regions, accelerating the workflow by **67%**.
- **Rebuilt logic gates** by resolving CNN/OpenCV mask discrepancies across Metal, Poly, OD, and Via layers.
- Converted model predictions into GDSII for netlist extraction and **KLayout LVS**, achieving **95%+** accuracy.
- Guided active-learning using **TTA** uncertainty, boosting **IoU to 93%+** and unseen-chip accuracy to **85%+**.
- Authored **illustrated documentation** for all projects to ensure seamless handoff to future team.

Java and C++ Programming Mentor

Sep. 2024 - Jun. 2025

Bur Oak Secondary School

Markham, Ontario

- Mentored students in **OOP** architecture, reinforcing design principles and problem-solving skills.
- Tutored C++ for NOIP, USACO, CCC, covering **DP**, **linked lists**, **Dynamic Array**, and **Hash Map**.

PROJECTS

RAG-Based IEEE Paper Assistant | Python, FAISS, LangChain, Gemini 2.0 API (LLM), PyMuPDF

An LLM-powered research assistant that turns dense academic papers into grounded, retrievable answers.

- Engineered a **RAG-based** research assistant using PyMuPDF to parse and clean double-column layouts.
- Built a vector database using **FAISS** and **Sentence-Transformers** for semantic retrieval of documents.
- Integrated LLM-generated summaries via **LangChain**, cutting document comprehension time to 3 minutes.

SEM-to-GDSII IC Layout Reconstruction | PyTorch, NumPy, Cadence Virtuoso

A PyTorch U-Net pipeline that converts SEM chip images into Cadence Virtuoso-compatible GDSII layouts.

- Achieved **95%+** IoU on subtle metal layers by developing a multi-channel **PyTorch U-Net CNN**.
- Reduced cross-circuit false positives by 30% using **CLAHE**, Sobel edge filters, and histogram matching.
- Extracted thin-line PO features from etched OD artifacts, improving layer accuracy from **<10% to 90%+**.

Microscopic Images Processing Suite | OpenCV, Python, Bash, KLayout, Virtuoso

An interactive computer vision application that transforms noisy microscopy to clear color-coded layer masks.

- Engineered a computer vision pipeline converting and fixing vague SEM images into 2-channel layer masks.
- Empowered non-technical users by engineering a **Tkinter GUI** for real-time parameter tuning and previews.
- Designed modules to correct SEM-specific image defects (noise, contrast drift, artifacts).

Touchscreen Image Editor | Kotlin, Java

An Android tablet app that lets users manually correct computer-processed images against their originals.

- Supported unbounded inputs by downsampling editing layers while preserving full-resolution lossless exports.
- Engineered stylus annotation canvas with **custom drawing tools**, reducing layer correction time by **67%**.

EDUCATION

University of Waterloo

Waterloo, Ontario

2A Bachelor of Applied Science in Computer Engineering - GPA 3.8/4.0

Sep. 2025 – Expected Apr. 2030

Relevant Courses: Algorithms and Data Structures, Digital Circuits and Systems, Analogue electronics